

Act I: You Don't Have To Live Next To Me

Act II: Data for

Act III: Magic Happens

Act I: You Don't Have To Live Next To Me

Towards Demobilizing Individualistic Bias in Computational Approaches to Urban Segregation

Anastassia Vybornova & Trivik Verma

Artwork by Namrata Narendra

<https://arxiv.org/abs/2505.01830>

Remember the 1960ies in the US?

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LOOK At These Homes NOW!

An entire block ruined by negro invasion. Every house marked "X" now occupied by negroes. ACTUAL PHOTOGRAPH OF 4300 WEST BELLE PLACE.

SAVE YOUR HOME! VOTE FOR SEGREGATION!



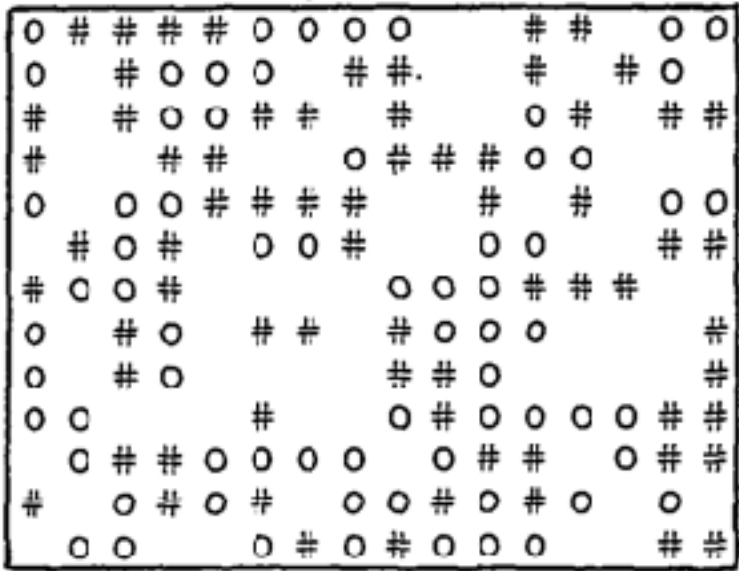
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MODELS OF SEGREGATION
By THOMAS C. SCHELLING
Harvard University



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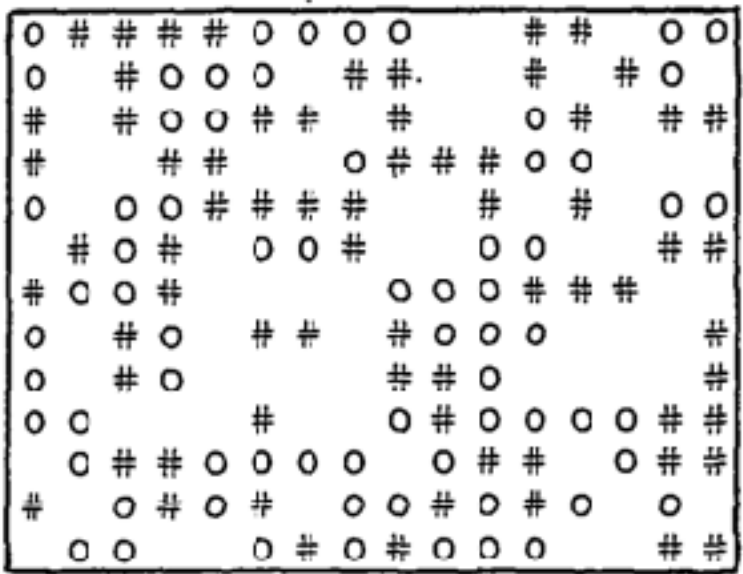


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Schelling (1969) <https://www.jstor.org/stable/1823701>

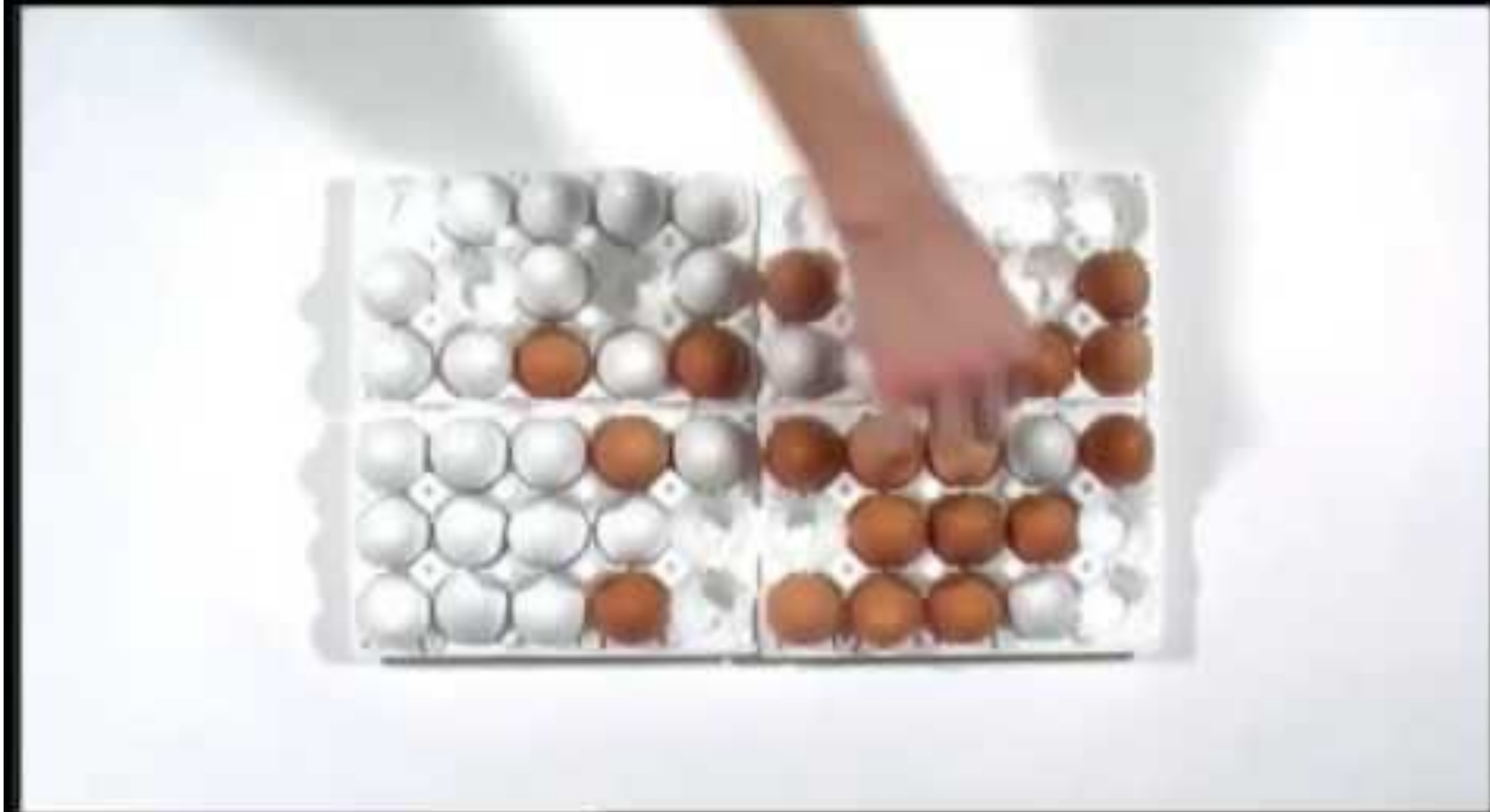
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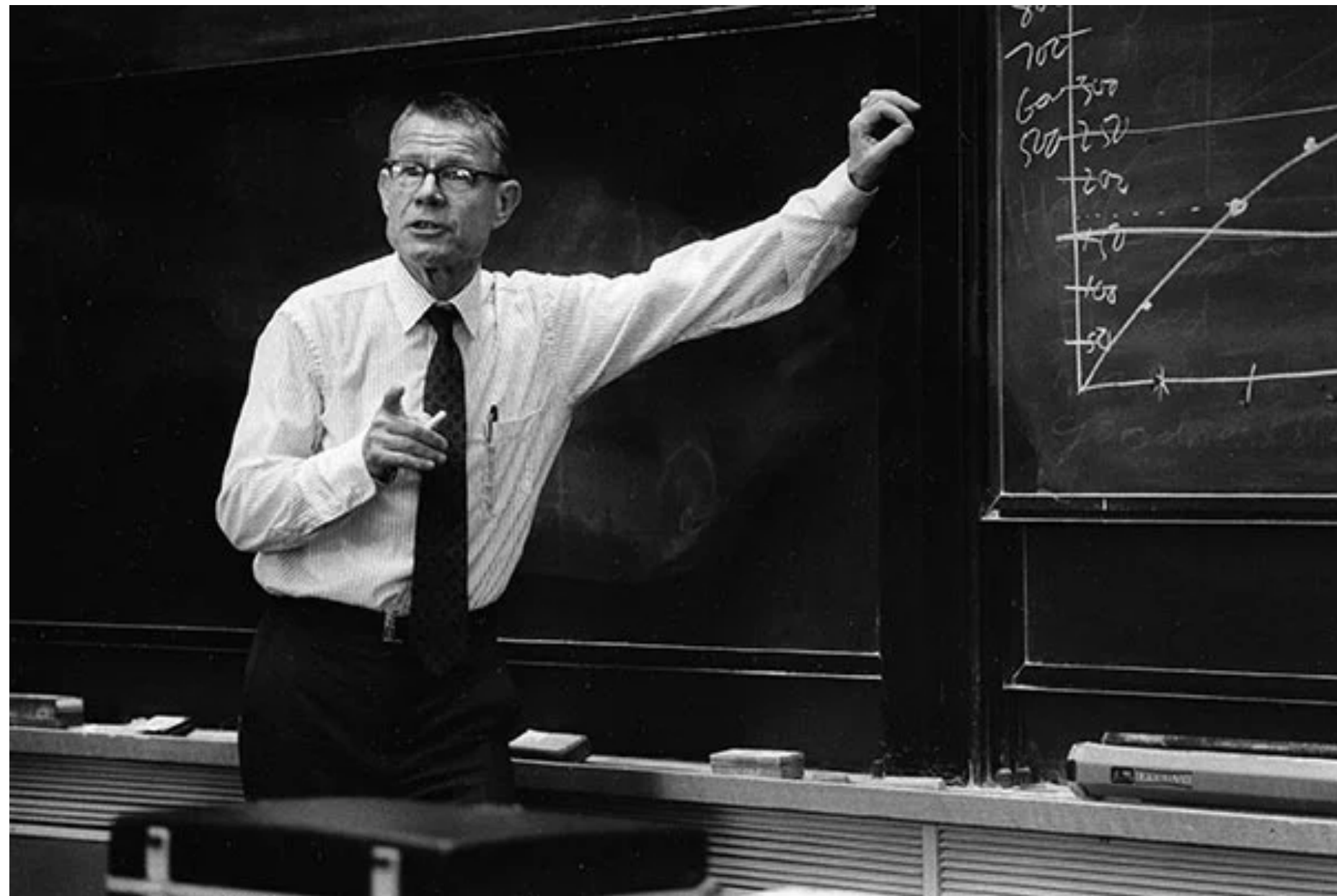
My ultimate concern of course is segregation by color in the United States; but at the level of abstraction of this paper, any twofold distinction could constitute an interpretation—whites and blacks, boys and girls, officers and enlisted men. The only requirement in my model is that the distinction be twofold, exhaustive, and recognizable.

Skin color, of course, is neither dichotomous nor even unidimensional; but by convention the distinction is twofold, even in the U.S. census.





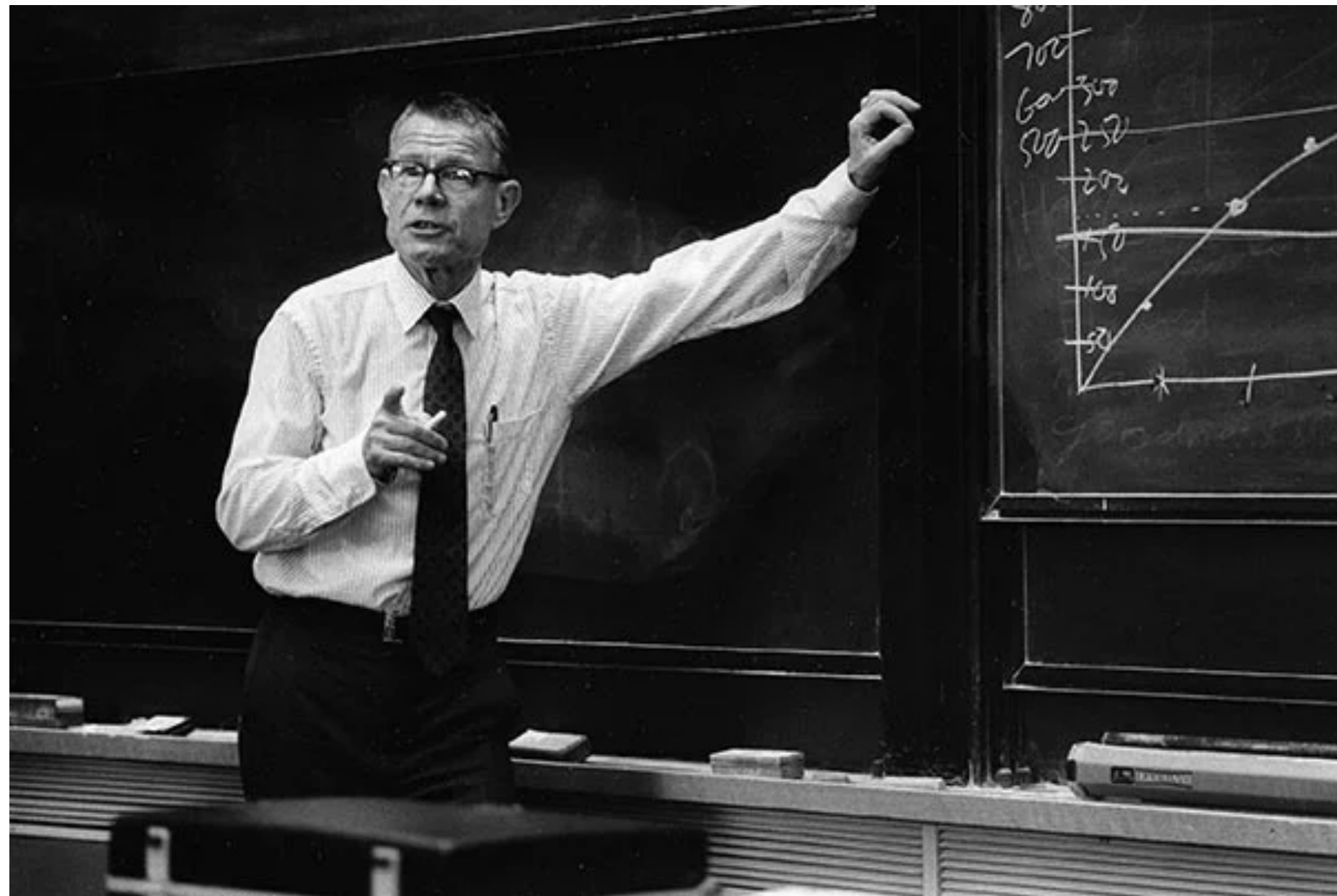
<https://youtube.com/watch?v=JjfihtGefxk>



Processes that lead to segregation,
by order of importance:

1. Governmental action (civil rights!)
2. Class differences (social equity!)
3. Individual action

(c) Schelling 1971



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But this was such a long time ago!

What has happened in the last 50+ years?



Fifty years after the Schelling's Models of Segregation: Bibliometric analysis of the legacy of Schelling and the future directions of segregation research

Rūta Ubarevičienė^{a,b,c,*}, Maarten van Ham^a, Tiit Tammaru^{b,d}

^a *Department of Urbanism, Delft University of Technology, Delft, the Netherlands*

^b *Department of Geography, University of Tartu, Tartu, Estonia*

^c *Department of Regional and Urban Studies, Institute of Sociology, Lithuanian Centre for Social Sciences, Vilnius, Lithuania*

^d *Estonian Academy of Sciences, Estonia*

It has been more than 50 years since Thomas Schelling (1921–2016) developed theoretical models explaining the dynamics of residential segregation. Today, the Schelling models of residential segregation have become one of the most cited models in the urban research literature.

explains clustering and separating, the most important issue in the Schelling model, from a social perspective, is how choices play out in the social fabric and lead to segregated residential patterns. We show here that there are well articulated mathematical explanations for social segregation, that simulation compared with having utility zero. Schelling used simple simulations based on such hypothetical preference schedules to show that the adjustments of individual households responding to changes in composition on the checkerboard invariably lead to complete segregation (3). New survey evidence for

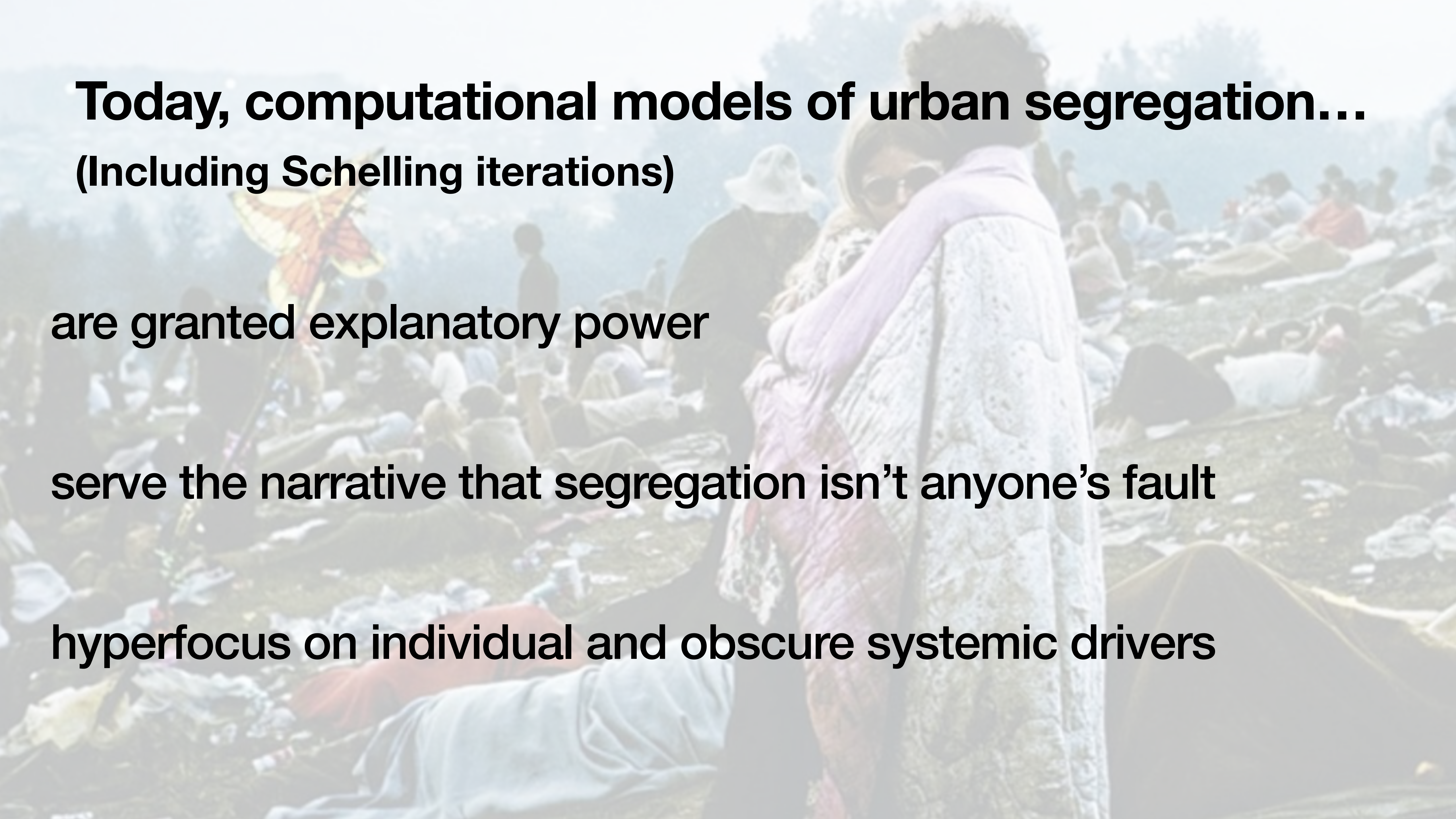
Clark & Fossett (2008) <https://www.jstor.org/stable/25461379>

One of the most striking examples of this is residential segregation (Sakoda 1971; Schelling 1971), where subtle behavioural responses at a microlevel are well known to have the potential to lead to severe consequences such as ethnic segregation at the macrolevel. A good understanding of individual

Liebe et al. (2023) doi:10.1177/00491241231186657

The Schelling model in its current form was published in 1971 by Thomas Schelling in a bid to understand how individual decisions of agents to relocate could lead to a macro pattern of segregation [10]. In the context of widespread racial segregation in the US, the model could demonstrate that for segregation to occur on a macro scale, no deeply entrenched racism was required- even

Urselmans (2018) doi:10.1007/978-3-319-90418-4_1



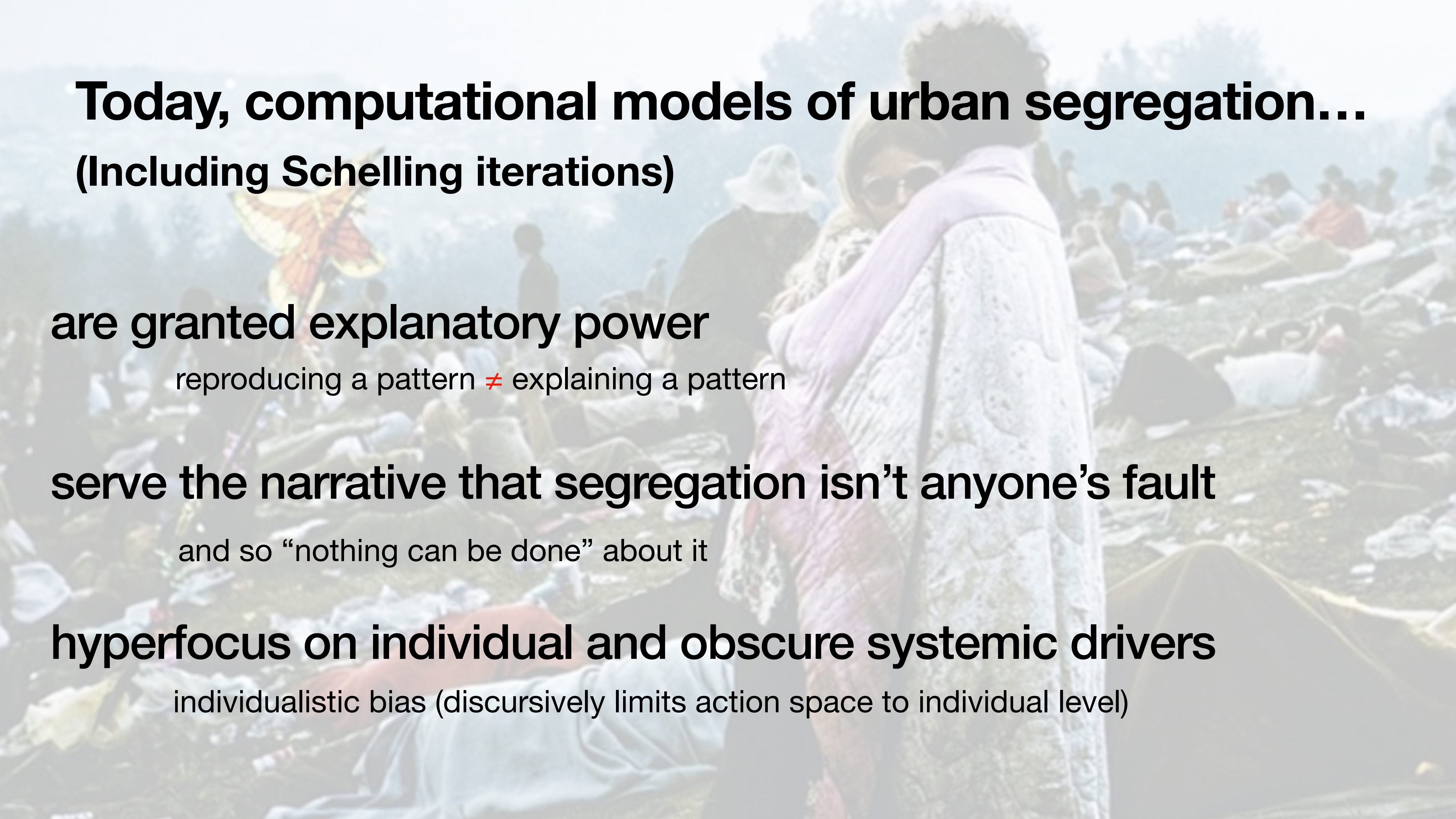
Today, computational models of urban segregation...

(Including Schelling iterations)

are granted explanatory power

serve the narrative that segregation isn't anyone's fault

hyperfocus on individual and obscure systemic drivers



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Framing **racism** as moral shortcoming of individuals....

“Systems of domination of any kind (...) do not depend for their reproduction on the actions of “bad actors”, that is, on the behavior of the sexists, racists, elitists, and homophobes. They remain in place fundamentally because of the actions and inactions of many good, nice people. Systemic racism is then maintained not by the actions of the “racists”, but by the passive, habitual, and mostly neutral behavior of the majority of Whites (neutrality in the face of systemic inequality amounts to supporting the status quo).”

Eduardo Bonilla-Silva (2008): Racism without Racists

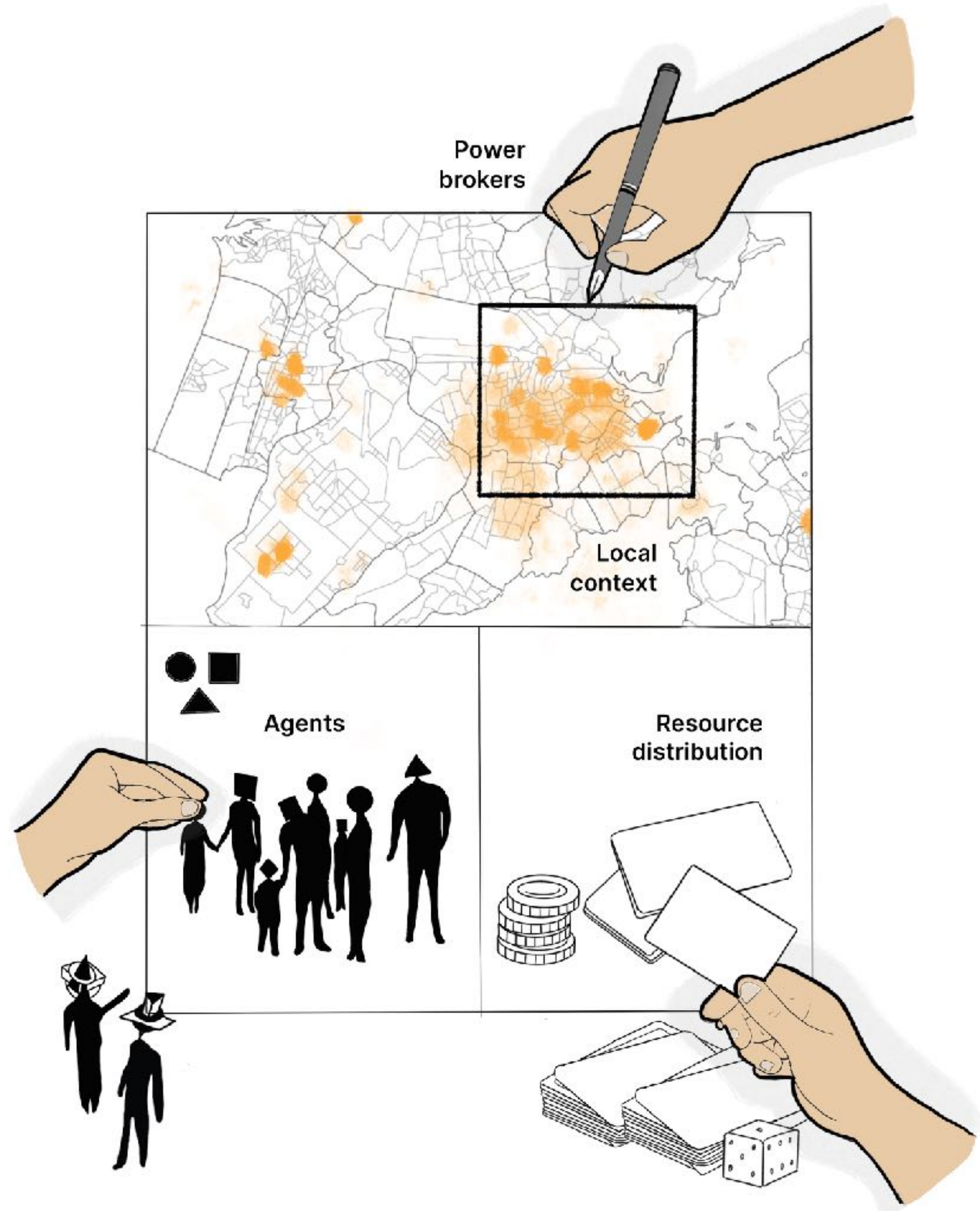
Cui bono?

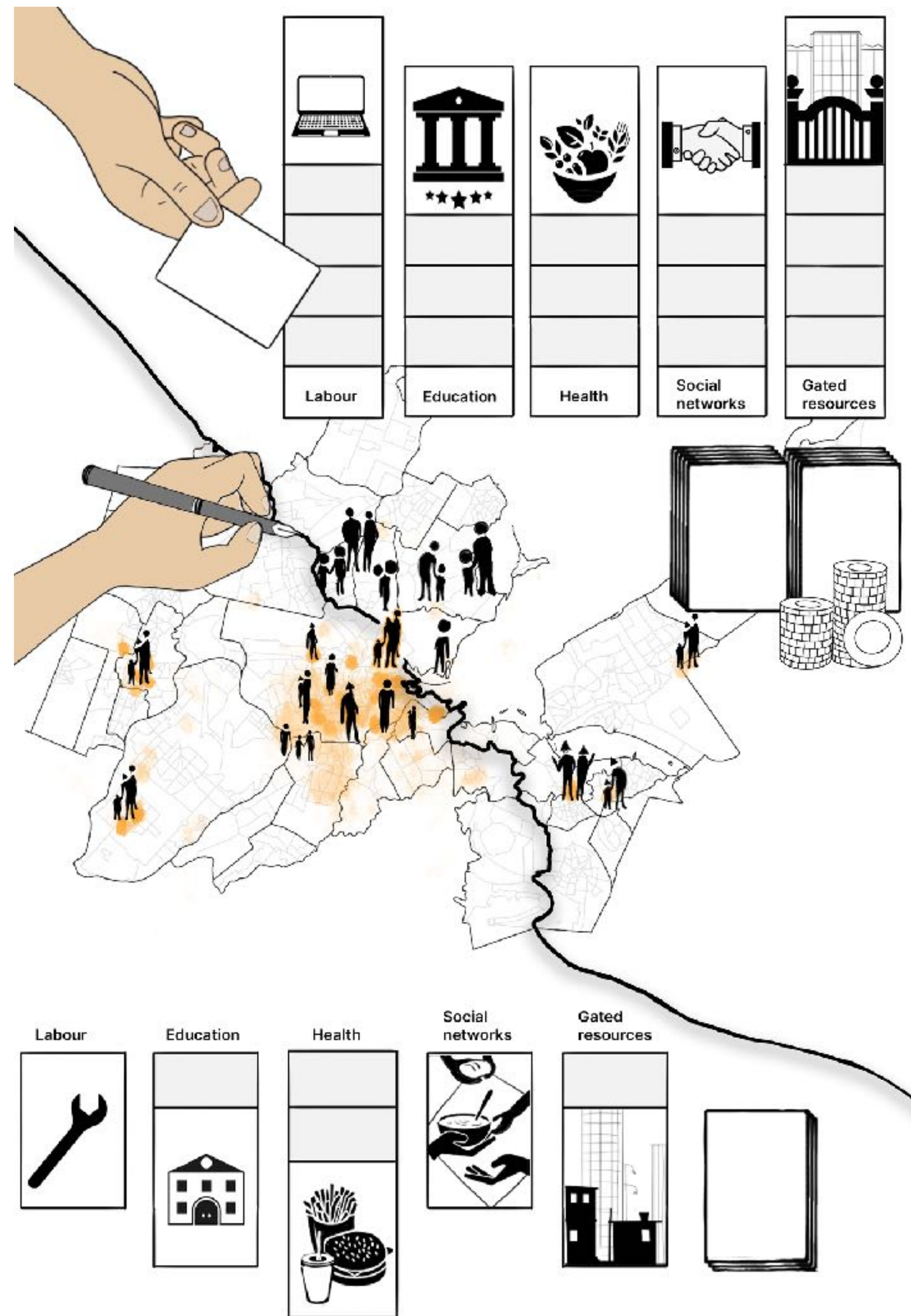
Framing **poverty** as moral shortcoming of individuals....

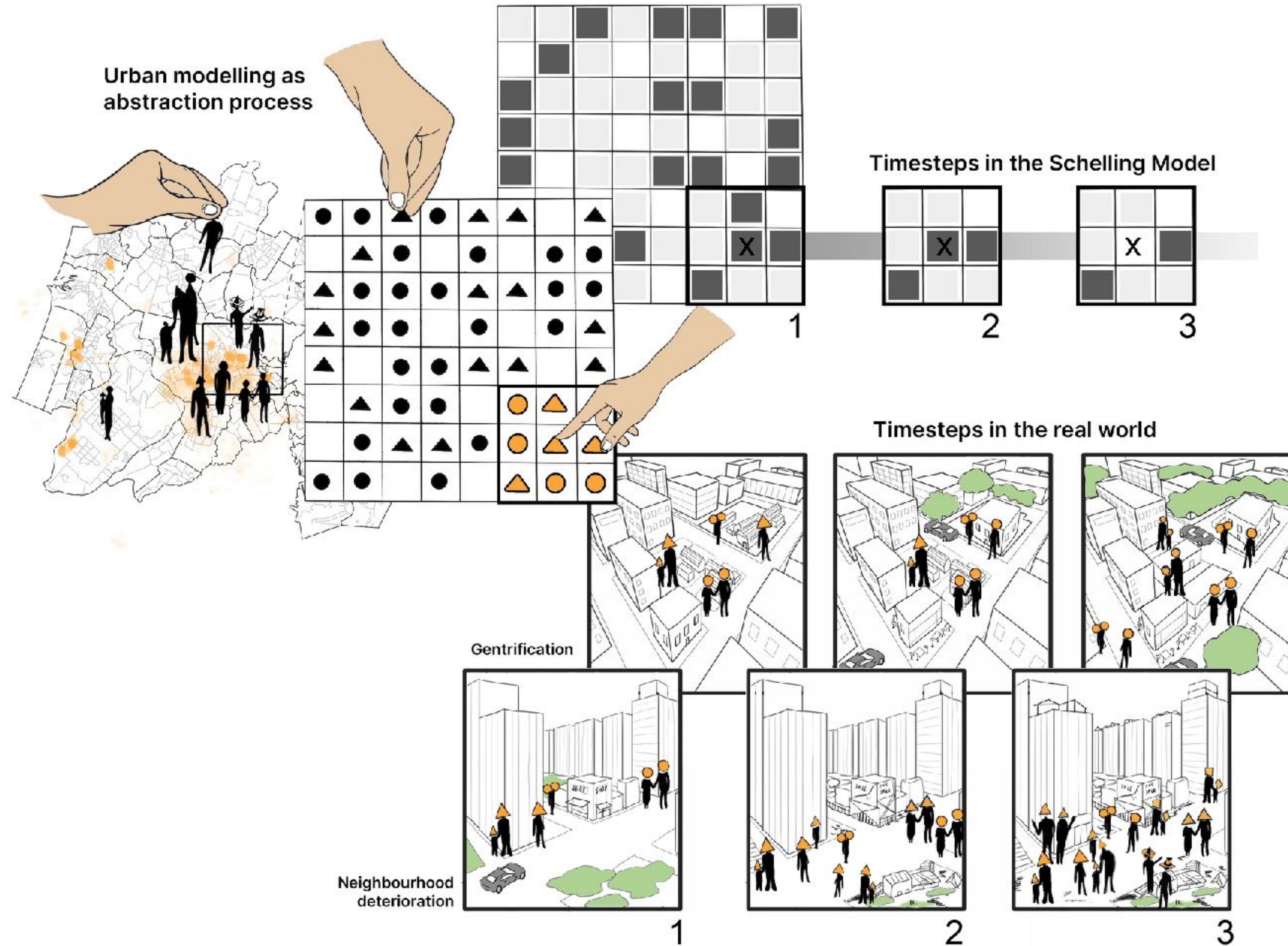
“(...) in 1987, an influential group of Dutch sociologists at the University of Leiden was developing a research program on ‘modern poverty’. This time period saw the beginning of the crisis of the welfare state and the gradual shift to a more neo-liberal regime. It was the start of a slow growth of acceptance of the perspective that holds the poor responsible for their poverty, or at least explains poverty by ‘a culture’ or personal choices and individual shortcomings.”

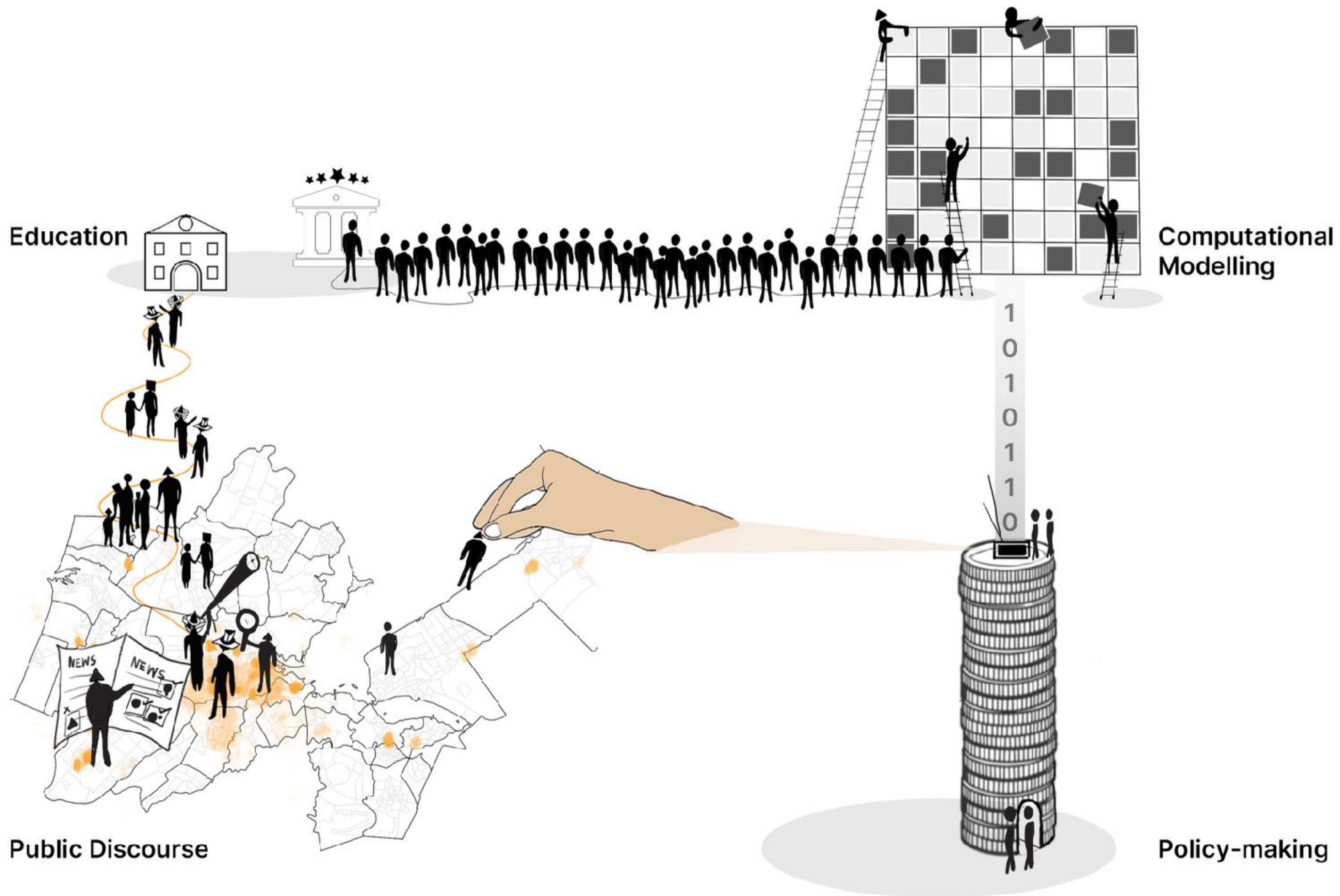
Tanja Blokland (2008) doi:10.1111/j.1540-6040.2008.00271_5.x

Cui bono?









Are You Western?

“Non-western foreign background”, as defined by Statistics Netherlands:

People from Turkey or countries in Africa, Latin America and Asia, with the exception of Former Dutch East Indies/Indonesia and Japan. The first generation consists of people born outside the Netherlands. The second generation are people born in the Netherlands of whom at least one parent was born abroad.

Due to their socioeconomic and cultural position, people from Indonesia and Japan residing in the Netherlands are considered as having a 'western' migration background. These are mainly people born in the former Dutch East Indies and expatriates employed by Japanese companies with their families.

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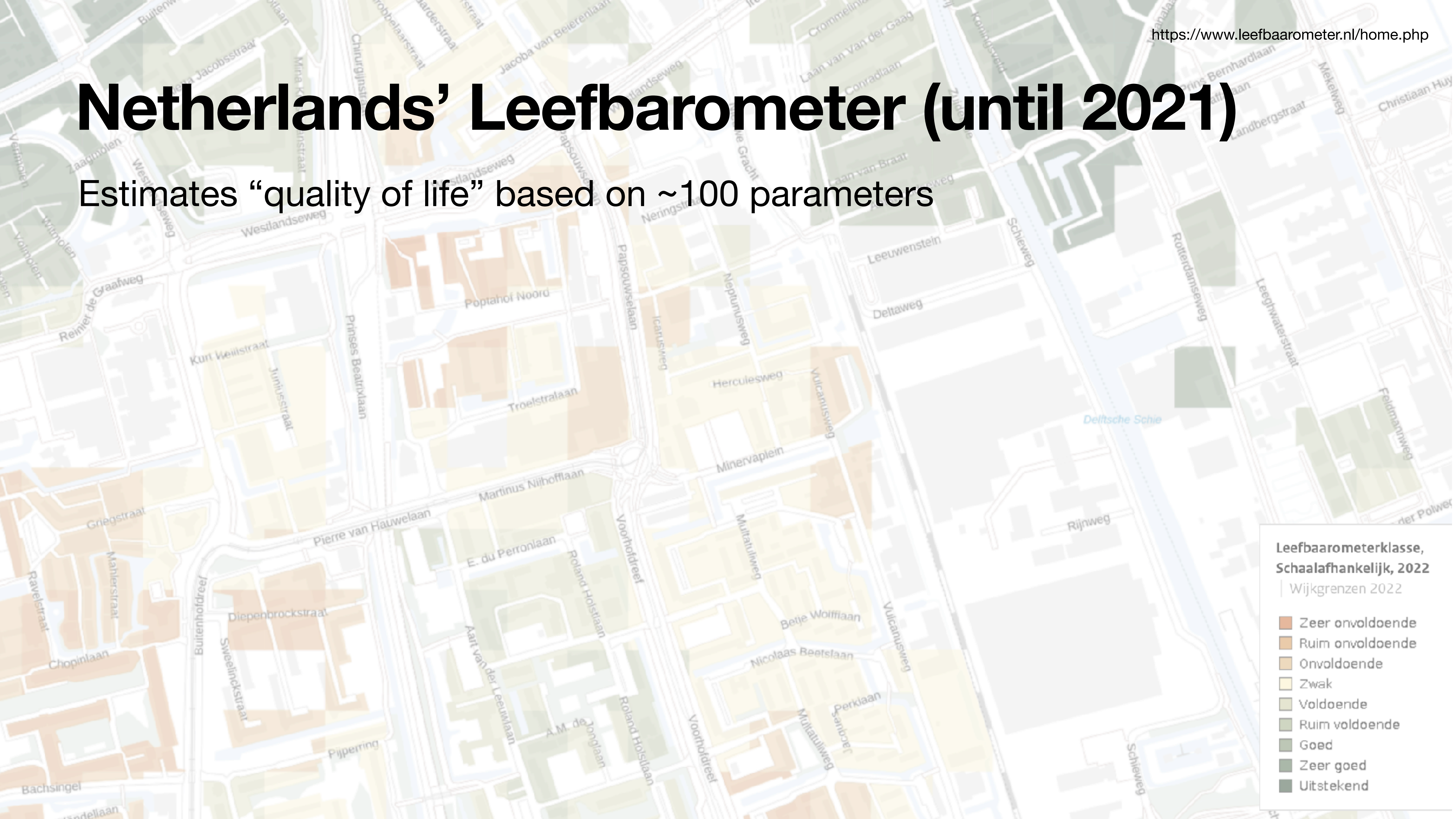
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Netherlands' Leefbaarometer (until 2021)

Estimates “quality of life” based on ~100 parameters



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Leefbaarometerklasse,
Schaalafhankelijk, 2022

| Wijkgrenzen 2022

- Zeer onvoldoende
- Ruim onvoldoende
- Onvoldoende
- Zwak
- Voldoende
- Ruim voldoende
- Goed
- Zeer goed
- Uitstekend

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+ “Western immigrants”

- “Non-Western immigrants”

- People who receive social benefits

- People who are looking for a job

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■	Zeer onvoldoende
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Denmark's 'Ghetto laws' **(rebranded as "Udsatte boligområder og parallelsamfund" in 2021)**



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Percentage of people from
“Non-Western” countries,
& their descendants, and:

- Unemployment rates
- Crime rates
- Education level
- Income level

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**Publishing a yearly
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**“The first stage of the
development project
is the demolition of
188 homes.”**

We need to demobilize individualistic bias in computational approaches to urban segregation

Nina Simone - Mississippi Goddamn



<https://www.youtube.com/watch?v=LJ25-U3jNWM>

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You don't have to live next to me

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**You don't have to live next to me
Just give me my equality**

Act II: Data for

is (...) a political system, a particular power structure of formal or informal rule, socioeconomic privilege, and norms for the differential distribution of material wealth and opportunities, benefits and burdens, rights and duties.

is the unnamed political system that has made the modern world what it is today.

Act II: Data for White Supremacy

White supremacy is (...) a political system, a particular power structure of formal or informal rule, socioeconomic privilege, and norms for the differential distribution of material wealth and opportunities, benefits and burdens, rights and duties.

White supremacy is the unnamed political system that has made the modern world what it is today.

Charles W. Mills, *The Racial Contract* (1997)

At the example of the so-called “parallel societies index”, which has been in use in Danish policy-making since 2018, here we show how policy-makers mobilize data and models to reify white supremacy.

The “parallel societies index” in Denmark

1. Living in a public housing area where there are >25% NWIAD
2. Children aged 1-4 at home
3. Children aged 0-6 in daycare where >25% NWIAD
4. Children aged 6-16 in school where >25% NWIAD
5. Convicted children aged 12-14
6. Convicted children and young adults aged 15-29
7. Children and young adults aged 16-29 welfare benefits for 6 months
8. Adults aged 22-59 welfare benefits for 4 years

***NWIAD: Non-Western Immigrants and their descendants**

The “parallel societies index” in Denmark

0. Being categorised as “Non-Western”

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Our models are reifications of individualistic bias.

A recurring issue in inequality discourse is the misalignment between philosophical positions of the stated future goal of eradicating inequalities and the proposed pathways to achieve it, often rooted in individualistic meritocracy or individual welfare.

This is the central predicament of the economy-engineering complex (EEC). **Not all people genuinely believe that all inequalities can - or should - be eliminated.** The deeper issue is not the validity of this philosophical stance (independently of whether or not we agree with it), but its implicit and unconscious character, as often seen within the EEC.

As Victoria Reyes points out, this predicament can be better understood when considering that the academy is overwhelmingly white, wealthy, male, and Western; as a result, the institution globally lacks the collective experience of oppression. This lack of experience feeds into the misalignment described above, producing an ideological lock-in.

People who work towards inequality reduction, while ignorant of their own disbelief in the feasibility of this goal, heavily constrain the solution space in which the economy-engineering complex operates.